UNIVERSITY OF WINDSOR

Faculty of Engineering

Department of Mechanical, Automotive and Materials Engineering

Probe Holder Design Overview

Submitted on Tuesday, August 25, 2026

Philipp Kutsyllo

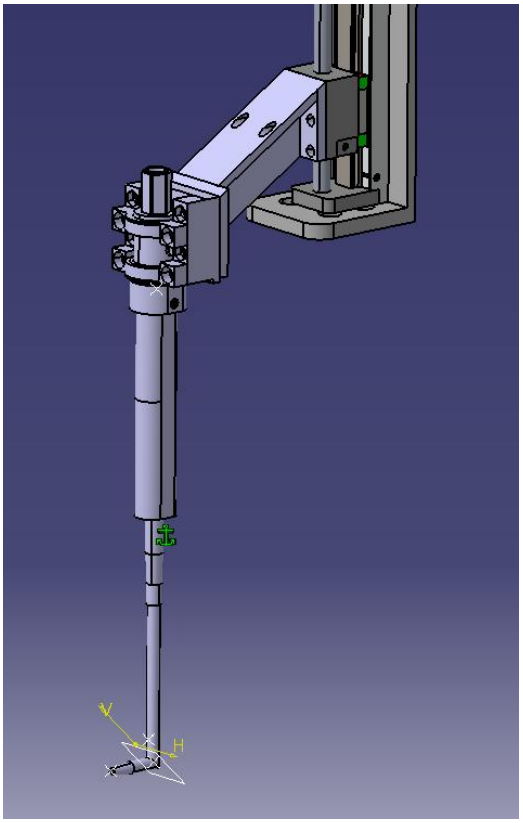
Student ID: 110124671

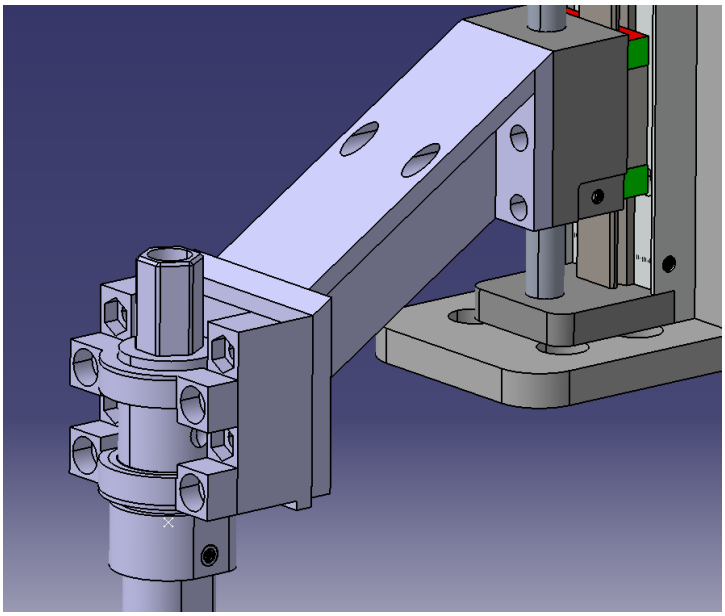
BOM

QTY	Part Number	McMaster Carr ID
2	M3 Brass Heat-Set Insert	N/A
2	M3x0.5 5mm Set Screw	N/A
2	Top Clamp	N/A
1	Plunger Holder	N/A
1	Press-Fit Ball-Nose Spring Plunger	4232N22
1	Brace	N/A
1	Probe Holder/Turner	N/A
8	M4x0.5 25mm Zinc-Plated Alloy Steel SHCS	90128A217
8	M4x0.5 Zinc-Plated Steel Hex Nut	90591A255

M3 Brass Heat-Set inserts *are* available at McMaster Carr under 94459A769, however, the CAD of the Probe Holder was sized for commonly available Amazon-sourced M3x5x4 inserts, such as these. A version sized for the McM-C inserts is included, “Probe holder-turner for McM-C insert.CATPart” (and .STEP).

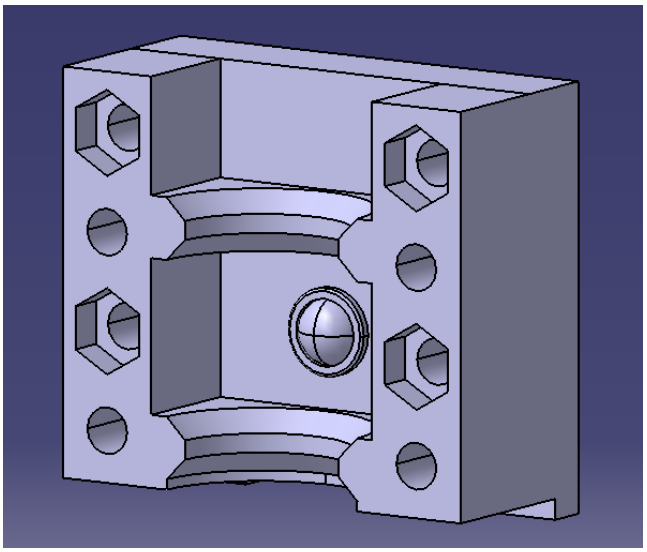
Design Overview



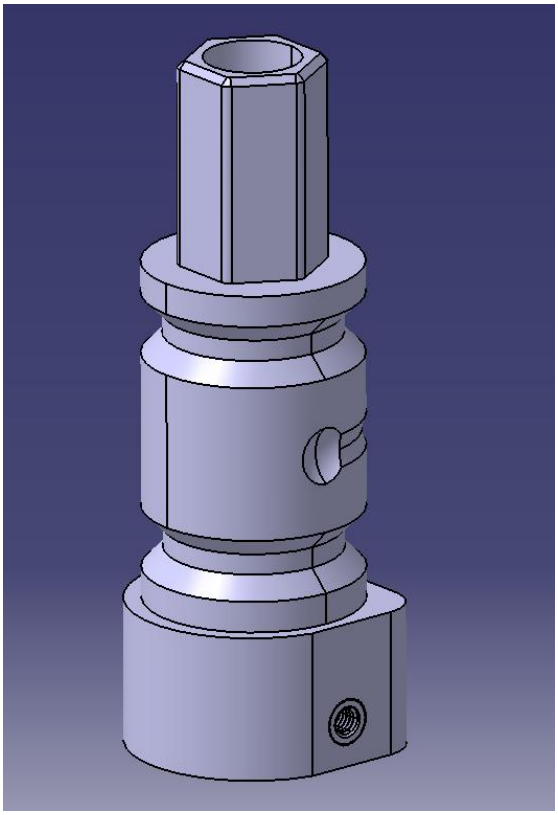


Assembly Manual

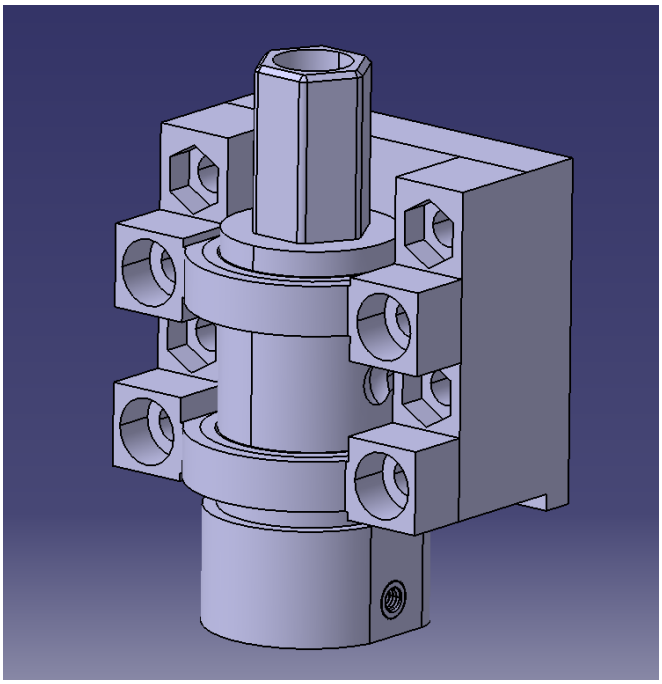
1. Press in all press-fit M4 nuts into the Plunger Holder (not pictured)
2. Press in the Plunger into the Plunger Holder



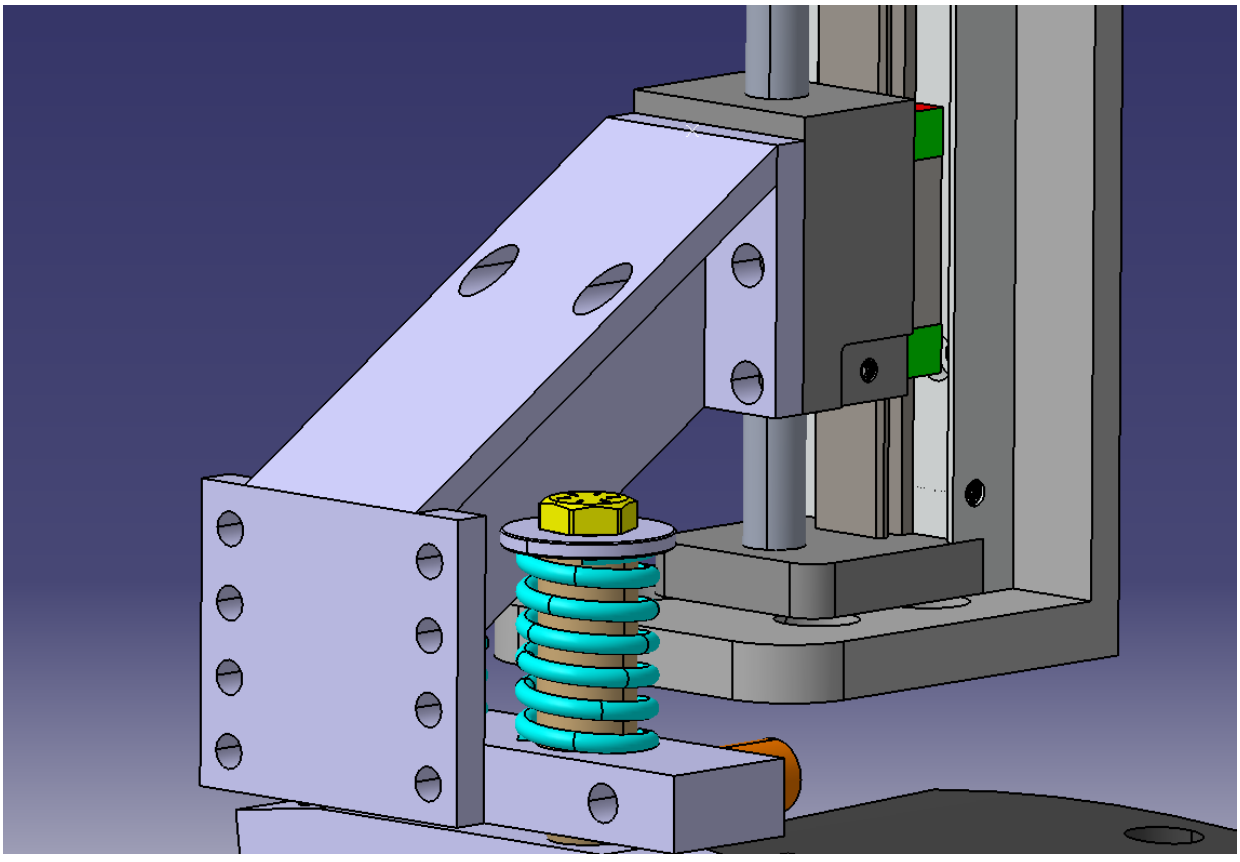
3. Heat-set the inserts into the Probe Holder with a soldering iron (tip: Rocketry Team has a good rig that ensures they are square). NOTE: most inserts are directional! If using a short (under 4mm) insert, insert with the probe extension in, to avoid overinsertion.



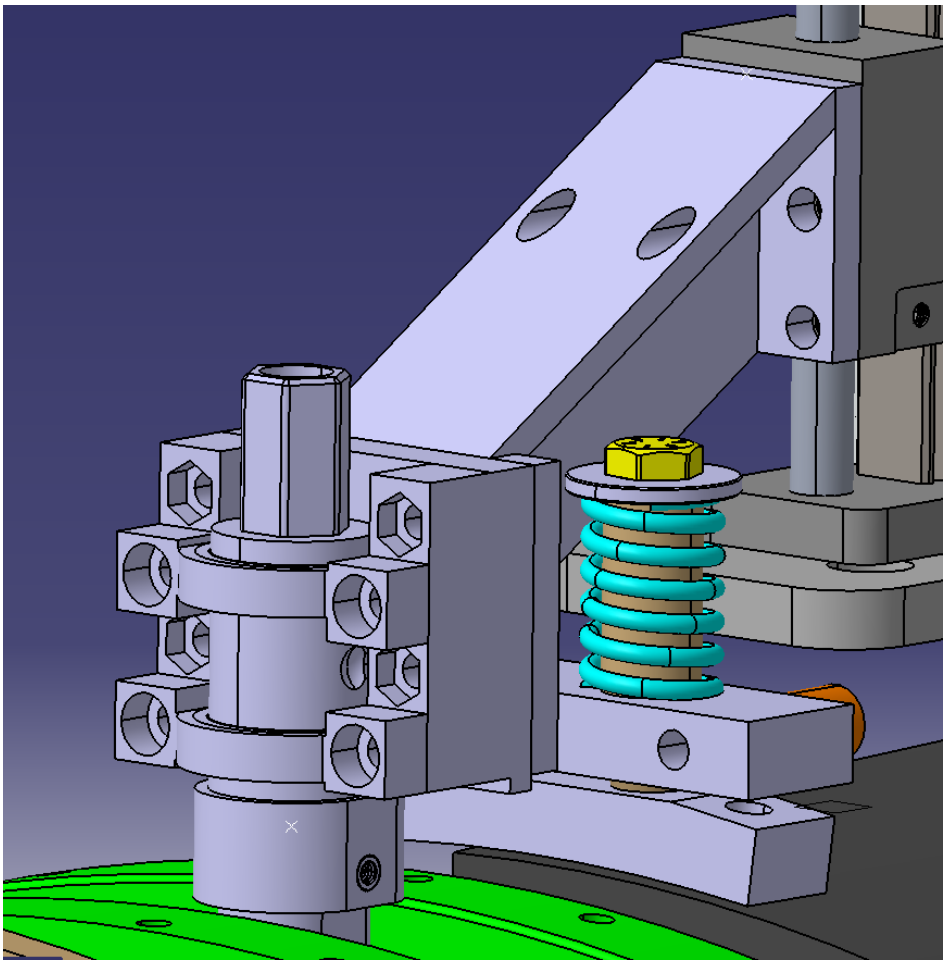
4. Put the Probe Holder and the Plunger Holder together, and clamp with the two Top Clamps. Screw in the fasteners (not pictured)



5. Attach the Brace to the Linear Slider (fasteners not pictured and not included in BOM)

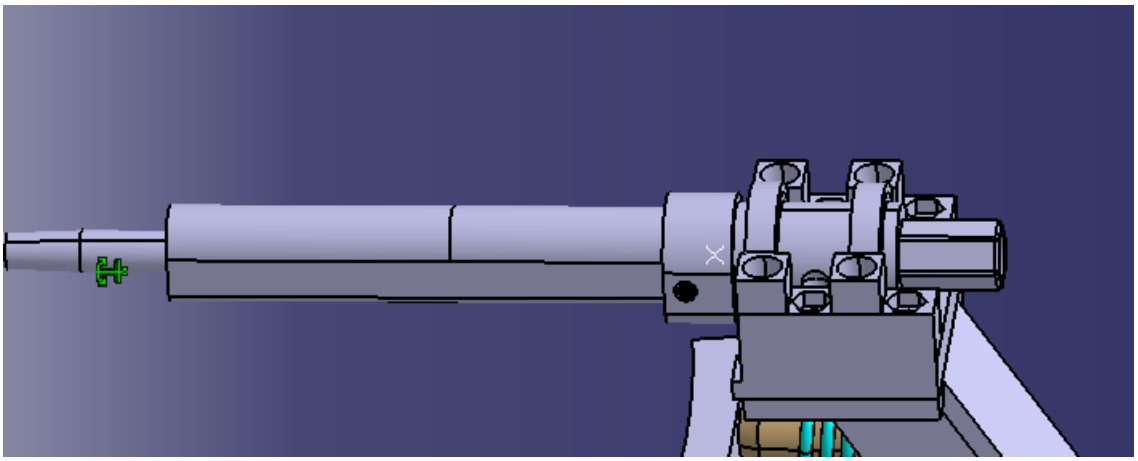


6. Attach the assembly to the Brace



Note the two holes provided for convenient hex key access

7. Attach the Probe Extension to the Probe Holder with the set screws (not pictured)



NOTE: Route the pitot tube extension tubes through the top before/during this step

Operating Manual

Use a ½-inch or 13-mm crescent wrench to turn the top of the holder. The spring-loaded ball detent will align the probe facing -90, 0 and 90 degrees.

Design considerations

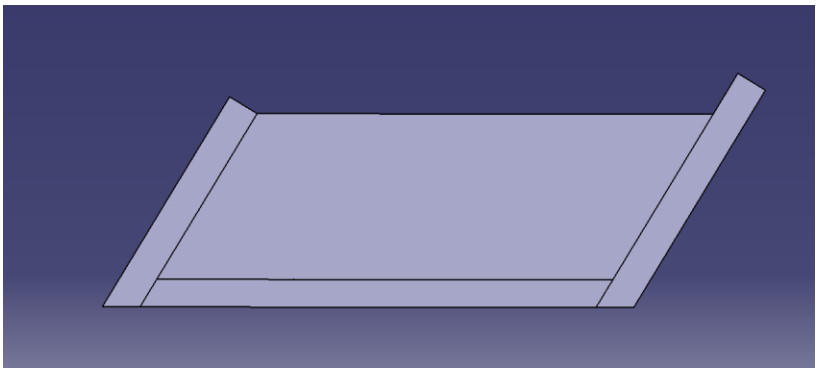
The probe holder is designed to resist loadings in all possible orientations on the rig. The rib on the Brace is intended to resist the bending loading present in the non-vertical orientations. All parts are designed to be 3D-printed with minimal supports. Initially, I wanted to use a lot less fasteners, but the strength and rigidity tradeoffs that came with that were unacceptable. The two clamps constrain the probe holder axially, while the ball detent constrains the rotation with three locked positions for easier probe removal.

More than five different designs were considered, with many being rejected for insufficient rigidity/excess complexity. The current design, while requiring a lot of fasteners, promises to be rigid and reliable, and easy to operate once assembled.

General Printing Advice and Favorable Orientations

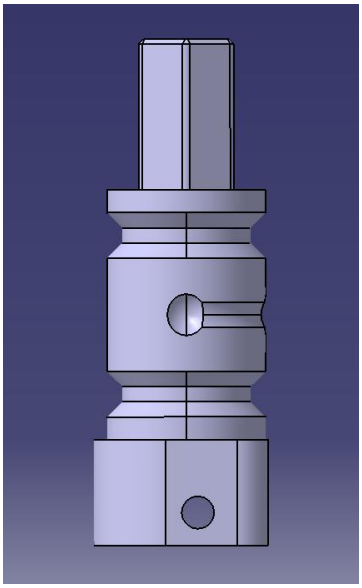
As discussed in-person previously, all parts are to be printed out of ABS. This filament can be annoying to work with, requiring high chamber and bed temperatures. Additionally, it is prone to shrinking. Some parts may need to be printed at 101…103% scale to ensure correct fits. Unfortunately, on this timeline I was not able to do physical testing in advance to implement the necessary corrections.

Brace



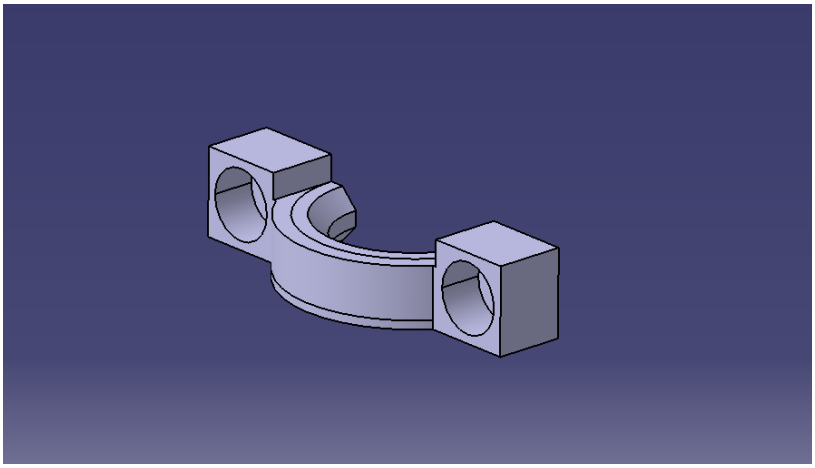
This orientation requires minimal supports, only in the tool clearance holes which are not critical

Probe Holder/Turner



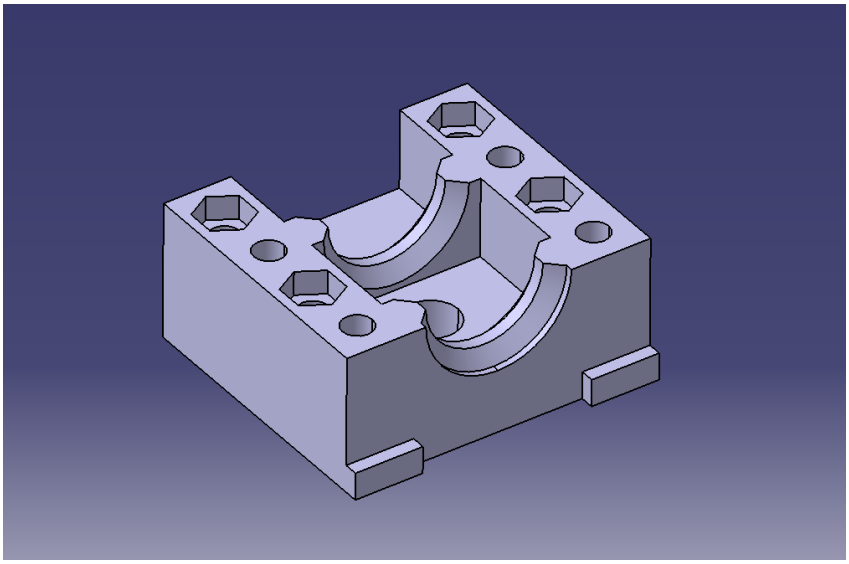
Note that if supports in the probe clamping area prove difficult to remove, an inverted print is possible with a brim to ensure bed adhesion.

Top Clamp



Note that supports will be necessary for the holes, but the contact surface quality is more important here.

Plunger Holder



Take care when removing supports from the bottom hex areas not to damage the side wall. Could be potentially printed supportless if bridging is good enough.

Closing Remarks

The setup requires physical testing after installation and before rig operation. Failure of the holder is not an option, so an evaluation in-situ is necessary to avoid catastrophic failure. Additionally, rigidity analysis is needed to ensure accuracy between the assumed and the actual probe tip location. I am available for any necessary revisions/redesigns at kutsyll@uwindsor.ca or over Teams.